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RESEARCH ARTICLE

IPL 2024: Unveiling the League Winner through Machine Learning Techniques

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ABSTRACT

In this dynamic unfolding of sports analytics, the core aspect of every groundbreaking research is to uncover hidden elements influencing future developments and results are labelled as sorcery. They can be manifold by a range of approaches. Nonetheless, in the dominated big data landscape Machine learning algorithms to discern patterns and produce prognostications in various fields, encompassing innumerable sports, especially Cricket is favoured globally and is no exception. In India, people distinctly rejoice in cricket. One among them is the T-20 format. This paper focuses on envisaging the 17th edition of the Indian Premier League (IPL) scheduled for 2024 in India. Among supervised techniques, we used a random forest algorithm model to predict the 17th edition of the franchise Twenty20 cricket league winner, which emerged as the most effective, achieving a custom accuracy of 64.5%. According to our analysis, the Kolkata Knight Riders, are foreseen to win the IPL 2024. We utilized a dataset from Kaggle for this study.

Keywords: machine learning, IPL 2024, winner prediction, Random forest algorithm

INTRODUCTION

In India one of the most prevalent and prominent sports is cricket. The viewership and popularity have increased tremendously over the decades. skimpy that cricket has more than two billion fans worldwide, with the potential for significant growth. Among all formats,T-20 commemorates its standard, in which the Indian Premier League (IPL) a men's (T20) cricket league is notorious. Every last one of these cricket fans is waiting with bated breath about upcoming cricket events and tournaments. They desire to cram about the prospects of their favorite team. Through this lens, analysis is directed to speculate on the 2024 league winner by leveraging machine learning techniques and historical data. The vigorous growth of machine learning and big data grants a distinctive chance to obtain and uncover significant insights from extensive data amassed by the IPL, incorporating other contributing factors. By harnessing this data, this study aspires to project the league winner of the 2024 series. The obstacle array is not only in treating this comprehensive dataset but also in building a model algorithm to navigate the outcome. Our strategy encompasses collecting and preprocessing intricate datasets, ensued by implementing a commonly used supervised machine learning algorithm (Random forest classifier)and aggregated technique to detect the repercussions with an accuracy score. This paper accentuates Feature Engineering, data splitting, model divination, and evaluation. Via this paper, the entire group of cricket fans can fathom ML algorithms integrated with sports prediction

Review of Literature

The literature on envisioning sports results, chiefly within the monarchy of cricket and exactly the Indian Premier League has grown significantly with the advent of machine learning and analytics techniques. Numerous research studies have ventured to model the end products of IPL matches using several statistical and machine-learning models emphasizing the broad scope of cricket analytics. A dominant feature of research prioritizes machine learning algorithms, with studies like Srikantaiah KC *et al*(2021) exploring the various factors that influence the outcome of IPL matches, among the algorithms used a random forest classifier is optimal. Likewise, G. Sudhamathy *et al*(2020) have cast off the R package by implementing four different algorithms to explore the dataset and represent the conclusions in a graphical representation. For instance, Nikhil Dhonge *et al*(2021) proposed a comparative analysis to give important information like IPL score prognosis by implementing three different regression models in which the linear regression model stands out, and for winning speculation among RFC, SVC, and DTC, RFC ML algorithms among linear, ridge, and lasso regression and for winning between SVC, Decision tree, RFC got more accuracy than others. In some cases, the SVM algorithm achieves the uppermost accuracy in recognizing the optimal attributes affecting the match outcomes in the investigation by PranavanSomaskandhan *et al*(2018).Moreover, constructing ML approaches to predict game results with a precession of 75% is good and was conferred by Sunil Bhutada *et al*(2020),further assessing the team's improved strength. Song. H et al(2011) have enhanced mobile node location estimation in cricket Sensor Networks by using FPB-IMM outperforming the Kalman filter and CIMM in accuracy and reliability. Priyanka. S *et al*(2020) have emphasized the match outcome using the Random Forest algorithm. Singh, D. K. *et al* (2021) have addressed the uncertainty in T-20 cricket by leveraging evolving technology. Multivariable linear regression algorithm, demonstrating higher accuracy through metrics such as MSE, RMSE, and R^2 score to forecast IPL match outcomes by Kavitha, P. V. *et al(*2021) Overall, the literature illustrates a shift between various erudite models that lecture the complexities of cricket by machine learning algorithms. For strategic decision-making, data-driven insights are increasingly essential in sports analytics.

PROPOSED METHODOLOGY

The methodology for unveiling the IPL league winner involves a reiterative process that take part in data collection, preprocessing, model selection, and validation. The fundamental objective is to implement a prophetic model that can accurately conjecture the potential winner of IPL 2024.

Data Collection

An all-inclusive dataset is gathered from Kaggle encompassing historical IPL match data from various seasons. This embraces complete statistics such as venues, toss decisions, team names, winners, and circumstantial factors. This dataset has extracted meaningful features that contribute to match results.

Data Preprocessing
The gathered data is imperilled to preprocessing to remove whitespace, handle missing values, and correct team names. Feature engineering plays a decisive role in this stage, label encoding is used to convert the string into a numerical value in the team column.

Model Selection
As from the referred paper, the RFC model gives optimal accuracy compared with other ML algorithms mentioned, and this paper is built using a random forest classifier algorithm predicting IPL league outcomes.

Training And Tuning
The dataset is split into training and testing sets based on the Season column. For the training stage matches up to the 2023 season are considered, while for testing twenty-24 season is undertaken. The chosen model is trained on a training dataset. The projected labels are mapped back to team names using a predefined mapping.

Model Validation
The predictive model is corroborated using a dataset not used during the training stage. Accuracy metrics are hired to appraise the model's performance are plotted.

Findings
The forecasted league winners of IPL 2024 are reported with accuracy scores. This methodology aims to provide a full-bodied and stead fast machine-learning framework for estimating the winner of a league, causal to the mounting field of sports analytics and enhancing our understanding of the factor that is causative to feat on T-20 cricket.

Visualization: Univariate Analysis
Fig 1reveals a horizontal bar chart that Mumbai Indians (MI) leads with a score of 133, followed closely by Chennai Super Kings (CSK) at 125 and Kolkata Knight Riders (KKR) at 124, indicating they are the top-performing teams. Royal Challengers Bangalore (RCB), Sunrisers Hyderabad (SRH), Rajasthan Royals (RR), Delhi Capitals (DC), and Pune Warriors (PK) fall into a mid-range performance bracket, with scores ranging from 96 to 108. In contrast, Lucknow Super Giants (LSG) and Gujarat Titans (GT) are significantly behind, with the lowest scores of 44 and 36 respectively, highlighting a notable disparity in performance levels among the teams. Fig 2 represents the top 10 cricket stadiums by the number of matches played. Wankhede Stadium tops the list with (111) matches, followed by Eden Gardens (77) and M Chinnaswamy Stadium (63). Other notable venues include Feroz Shah Kotla (59), MA Chidambaram Stadium (48), and Rajiv Gandhi International Stadium, Uppal (47). Sawai Mansingh Stadium also hosted 47 matches, while Punjab Cricket Association Stadium (34), Sheikh Zayed Stadium (29), and Maharashtra Cricket Association Stadium (22) round out the list. Wankhede Stadium stands out as the most frequently used venue. Fig 3 disclose Out of the total matches conducted from the first season to last season the decision spellbound each team after winning the toss,616 times chose to field first, and 345 times they chose to bat first. This shows that after the toss decision, most of the team prefers to field first

Prediction And Accuracy Evaluation
The Foreseen team, based on historical data and relevant features, suggests that has the highest likelihood of winning with a 64.48% accuracy score. However, the accuracy score indicates some uncertainty, and real-world outcomes may differ due to various factors.

CONCLUSION

Envisaging a winner in a field of sport is arduous and involves very convoluted methods, especially in cricket. but with the mystical arts of machine learning is easier. This work has an effect in predicting Kolkata Knight Riders as the champion of the 17th edition of IPL twenty- twenty-four using a Random Forest classifier to be the optimal achievement of a 64.24% accuracy score. As future scope to foreshow IPL league, as a hybrid approach incorporates player statistics, team strength, and historical data that could offer a comprehensive prophesied model. However, predicting a winner may not be correct in every situation this might change due to climatic conditions, auction reports, and player injury. consider social media sentiment to gauge fan support or use deep learning techniques to categorize patterns in historical match data.

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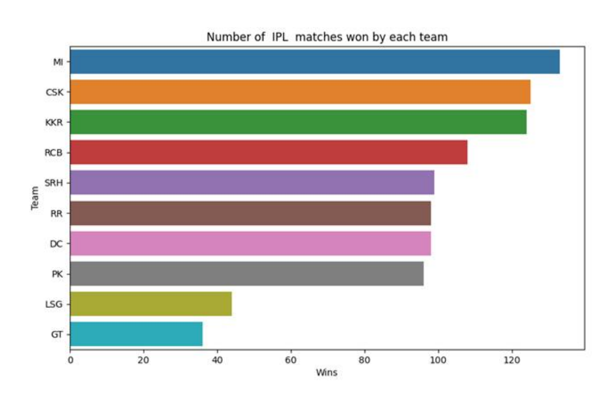

Figure.1: Number of Match Won By Each Team

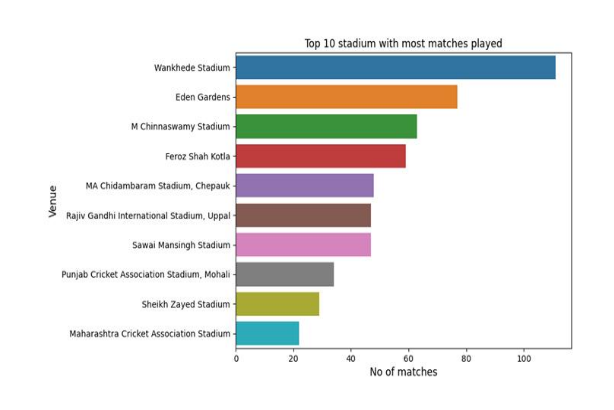

Figure.2: Top 10 Stadium With Most Matches Played

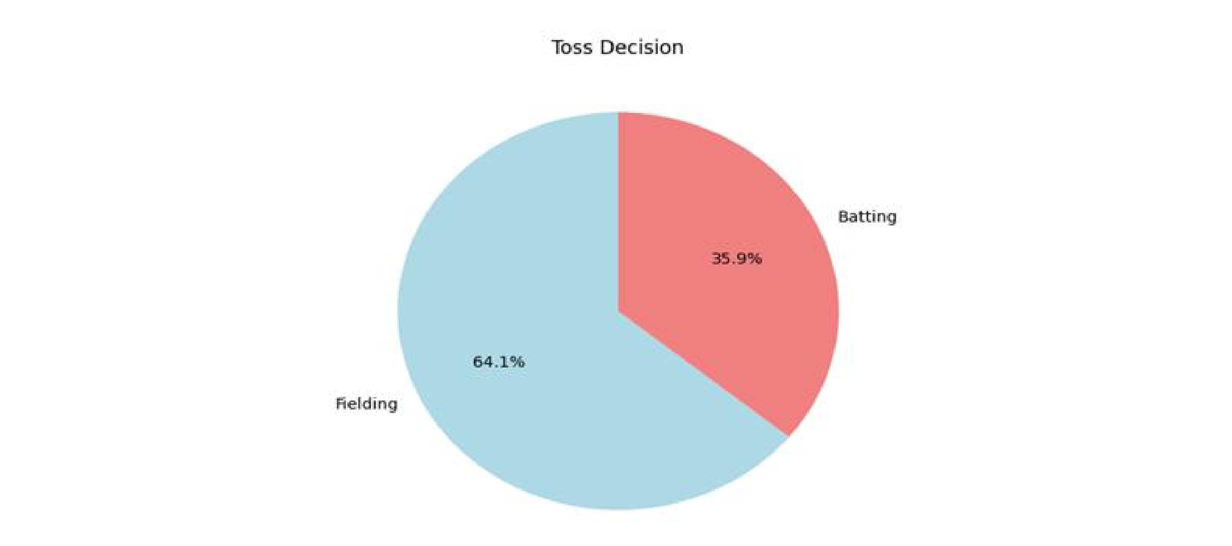

Figure.3: Toss Decision Taken By Each Team To Bat Or Fielding First